Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/06/2007
PI Name:	Singh, R Paul Ph.D.		
Project Title:	A Multipurpose Fruit and Vegetable Processing System for Advanced Life Support		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	95616-5294	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-01
Start Date:	01/13/2003	**End Date:**	01/12/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	3
No. of PhD Candidates:	4	**No. of Master' Degrees:**	12
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	12
No. of Bachelor's Candidates:	12	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

The following tasks were identified in the proposal:
1) Year 1 (January 1, 2003-December 31, 2003): a. Select varieties of tomatoes and other vegetables suitable for processing; b. Develop conceptual design for the Multipurpose Fruit and Vegetable Processor (MFVP); c. Develop computer-aided simulations of each operation to assist in design; d. Build bench-scale components of the MFVP and conduct preliminary trials for technical feasibility; e. Select optimum design parameters; f. Select membranes for water cleaning and product recovery.

2) Year 2 (January 1, 2004 – December 31, 2004)

a. Construct MFVP from components; b. Select membrane systems for concentration of tomato juice; c. Conduct trials with MFVP to determine operational feasibility; d. Determine equivalent system mass (ESM) and other operational

Task Description:	variables; e. Conduct optimization trials. 3) Year 3 (January 1, 2005 – December 31, 2005) a. Conduct experimental trials to determine safety, quality, and acceptability of foods processed in MFVP; b. Develop final modifications as indicated from the experimental data; c. Write final project report. To develop a miniaturized fruit and vegetable processor and to evaluate and optimize its performance for Advanced Life Support (ALS). The development of the fruit and vegetable processor represents the first attempt to miniaturize and integrate the array of unit operations required for preparation and processing of fruits and vegetables such as tomato. Additionally, the developers integrate the ESM metric evaluation throughout the development process as the primary project management tool.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research data on concentrating tomato juice using new generation membranes is valuable for the food industry. Only one Japanese company is currently employing this technique in a food plant located in California. Our data should prove useful for those food processors who are seeking optimization of the process.
Task Progress:	During the one-year no-cost extension, we completed the fabrication of the multipurpose fruit and vegetable food processor (MFVP) and the remaining trials with its components. In addition, we conducted a comprehensive study to obtain data on using electrolyzed water for cleaning-in-place (CIP) of the MFVP. CIP with electrolyzed water is an attractive option for cleaning as it does not require detergents (caustic solution) or disinfectants (chlorine). Therefore it could be more suitable for application in a controlled environment system. Electrolyzed water is generated from salt mixed in water. We conducted trials with a stainless steel test-bed containing various components used in the MFVP. The cleaning and disinfection of the components after inoculation was validated with microbial testing. Specifically, the effectiveness of electrolyzed oxidizing water (EOW) and electrolyzed reducing water (ERW) used in clean-in-place (CIP) applications for MFVP were studied using 38 mm (1.5 in) diameter stainless steel (SS316L) pipes fouled with apple juice. Data were compared with industrial CIP cleaner (chlorinated NaOH) and antimicrobial solution (mixture of peroxyacetic acid and hydrogen peroxide). Cleanliness of stainless steel (SS) pipe surfaces was quantified by a hygiene monitoring test and aerobic plate count. Data were obtained to determine the removal of organic debris and microorganisms at 5, 10, 15, and 20 min washing using electrolyzed water and industrial cleaning chemicals. Electrolyzed water showed potential use in CIP applications--it is easy to use and less hazardous compared to the industrial cleaning chemicals. Microbiological evaluation showed 5 min cleaning with EOW followed by 5 min cleaning with ERW was sufficient to drop cell levels from 5.2±0.3 log CFU (colony forming units)/stainless steel chip to below the limit of detection (<0.69 log CFU/cm2). In contrast, in case of bioluminescence tests, there were detectable adenosine triphosphate (ATP) even after 15 min of cleaning, even though the system was microbiologically at below detection. Furthermore, bioluminescence tests revealed that industrial cleaning chemicals were faster in cleaning compared to electrolyzed water. Electrolyzed water needs to be circulated more than 20 min at 0.6 m/s (2 ft/s) flow velocity to achieve acceptable cleaning, whereas industrial cleaning takes only 5 min of circulation for acceptable cleaning based on the bioluminescence readings.
Bibliography Type:	Description: (Last Updated: 03/06/2007)
Articles in Peer-reviewed Journals	Voit DC, Santos, MR, Singh RP. "Development of a multipurpose fruit and vegetable processor for a manned mission to Mars." Journal of Food Engineering. 2006 Nov;77(2):230-8. http://dx.doi.org/10.1016/j.jfoodeng.2005.06.035 , Nov-2006